CS 5990 - Assignment #3

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1. Binary classification problem
   - $P(+) = \frac{4}{9}$
     $P(-) = \frac{5}{9}$
     $Entropy = -\frac{4}{9}\log_2\frac{4}{9} - \frac{5}{9}\log_2\frac{5}{9} = 0.991$

   - $S = [4+, 5-]$
     - a_1:
       - $a_{1_{True}} = [3+, 1-]$
       - $a_{1_{False}} = [1+, 4-]$
       - $Entropy(a_{1_{True}}) = -\frac{3}{4}\log_2\frac{3}{4} - \frac{1}{4}\log_2\frac{1}{4} = 0.811$
       - $Entropy(a_{1_{False}}) = -\frac{1}{5}\log_2\frac{1}{5} - \frac{4}{5}\log_2\frac{4}{5} = 0.722$
       - $Gain(S, a_1) = 0.991 - \frac{4}{9}(0.811) - \frac{5}{9}(0.722) = 0.23$
     - a_2:
       - $a_{2_{True}} = [2+, 3-]$
       - $a_{2_{False}} = [2+, 2-]$
       - $Entropy(a_{2_{True}}) = -\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5} = 0.971$
       - $Entropy(a_{2_{False}}) = -\frac{2}{4}\log_2\frac{2}{4} - \frac{2}{4}\log_2\frac{2}{4} = 1$
       - $Gain(S, a_2) = 0.991 - \frac{5}{9}(0.971) - \frac{4}{9}(1) = 0.007$

   - Information gain for a_3
     - 0.5_{Split}:
       - $Entropy(\leq 0.5) = 0$
       - $Entropy(> 0.5) = -\frac{4}{9}\log_2\frac{4}{9} - \frac{5}{9}\log_2\frac{5}{9} = 0.991$
       - $Gain(0.5_{Split}) = 0.991 - 0.991 = 0$
     - 2_{Split}:
       - $Entropy(\leq 2) = 0$
       - $Entropy(> 2) = -\frac{3}{8}\log_2\frac{3}{8} - \frac{5}{8}\log_2\frac{5}{8} = 0.954$

- $Gain(2_{Split}) = 0.991 - \frac{8}{9}(0.954) = 0.037$

- 3.5_{Split}:
    - $Entropy(\leq 3.5) = 1$
    - $Entropy(> 3.5) = -\frac{3}{7}\log_2\frac{3}{7} - \frac{4}{7}\log_2\frac{4}{7} = 0.985$
    - $Gain(3.5_{Split}) = 0.991 - \frac{2}{9}(1) - \frac{7}{9}(0.985) = 0.0026$
- 4.5_{Split}:
    - $Entropy(\leq 4.5) = -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3} = 0.918$
    - $Entropy(> 4.5) = -\frac{2}{6}\log_2\frac{2}{6} - \frac{4}{6}\log_2\frac{4}{6} = 0.918$
    - $Gain(4.5_{Split}) = 0.991 - \frac{3}{9}(0.918) - \frac{6}{9}(0.918) = 0.073$
- 5.5_{Split}:
    - $Entropy(\leq 5.5) = -\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5} = 0.971$
    - $Entropy(> 5.5) = 1$
    - $Gain(5.5_{Split}) = 0.991 - \frac{5}{9}(0.971) - \frac{4}{9}(1) = 0.007$
- 6.5_{Split}:
    - $Entropy(\leq 6.5) = 1$
    - $Entropy(> 6.5) = -\frac{1}{3}\log_2\frac{1}{3} - \frac{2}{3}\log_2\frac{2}{3} = 0.918$
    - $Gain(6.5_{Split}) = 0.991 - \frac{6}{9}(1) - \frac{3}{9}(0.918) = 0.018$
- 7.5_{Split}:
    - $Entropy(\leq 7.5) = 1$
    - $Entropy(> 7.5) = 0$
    - $Gain(7.5_{Split}) = 0.991 - \frac{8}{9}(1) = 0.102$
- 8.5_{Split}:
    - $Entropy(\leq 8.5) = 0.991$
    - $Entropy(> 8.5) = 0$
    - $Gain(8.5_{Split}) = 0.991 - 0.991 = 0$

Class	+			-		+		-		+		-	+	-		
	a3															
	1			**3**		**4**		**5**		**6**		**7**		**8**		
	0.5		2		3.5		4.5		5.5		6.5		7.5		8.5	
	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>
+	0	4	1	3	1	3	2	2	2	2	3	1	4	0	4	0
-	0	5	0	5	1	4	1	4	3	2	3	2	4	1	5	0
Entropy	0	0.991	0	0.954	1	0.985	0.918	0.918	0.971	1	1	0.918	1	0	0.991	0
Gain	0		0.037		0.0026		0.073		0.007		0.018		0.102		0	

- According to information gain, the best split is a_1.

- Gain ratio:

	a_1	
	T	F
+	3	1
-	1	4

$$Gain\ Ratio(a_1) = 0.23/(-\frac{4}{9}\log_2\frac{4}{9} - \frac{5}{9}\log_2\frac{5}{9}) = \frac{0.23}{0.991} = 0.232$$

	a_2	
	T	F
+	2	2
-	3	2

$$Gain\ Ratio(a_2) = 0.007/(-\frac{5}{9}\log_2\frac{5}{9} - \frac{4}{9}\log_2\frac{4}{9}) = \frac{0.007}{0.991} = 0.007$$

	a_3	
	<=7.5	>7.5
+	4	0
-	4	1

$$Gain\ Ratio(a_3) = 0.102/(-\frac{8}{9}\log_2\frac{8}{9} - \frac{1}{9}\log_2\frac{1}{9}) = \frac{0.102}{0.503} = 0.203$$

According to gain ratio, the best split is a_1.

- Gini index of a_1 and a_2

$$GINI\left(a_{1_{True}}\right) = 1 - \left(\frac{3}{4}\right)^2 - \left(\frac{1}{4}\right)^2 = 0.375$$

$$GINI\left(a_{2_{False}}\right) = 1 - \left(\frac{1}{5}\right)^2 - \left(\frac{4}{5}\right)^2 = 0.32$$

$$GINI_{Split} = \frac{4}{9}(0.375) + \frac{5}{9}(0.32) = 0.344$$

$$Gain_{Split} = 0.5 - 0.344 = 0.156$$

$$GINI\left(a_{2_{True}}\right) = 1 - \left(\frac{2}{5}\right)^2 - \left(\frac{3}{5}\right)^2 = 0.48$$

$$GINI\left(a_{2_{False}}\right) = 1 - \left(\frac{2}{4}\right)^2 - \left(\frac{2}{4}\right)^2 = 0.5$$

$$GINI_{Split} = \frac{5}{9}(0.48) + \frac{4}{9}(0.5) = 0.4889$$

$$Gain_{Split} = 0.5 - 0.4889 = 0.011$$

- Classification error
  - a_1
    - $Classification\ error\left(a_{1_{True}}\right) = 1 - \frac{3}{4} = \frac{1}{4}$
    - $Classification\ error\left(a_{1_{False}}\right) = 1 - \frac{4}{5} = \frac{1}{5}$
    - $Classification\ error(a_1) = \frac{4}{9}\left(\frac{1}{4}\right) + \frac{5}{9}\left(\frac{1}{5}\right) = \frac{2}{9} = 0.222$
  - a_2
    - $Classification\ error(a_2) = \frac{2}{9} + \frac{2}{9} = \frac{4}{9} = 0.444$
  - Gain
    - $Classification\ errror(parent) = 1 - \frac{5}{9} = \frac{4}{9} = 0.444$
    - $Gain(a_1) = 0.444 - 0.222 = 0.222$
    - $Gain(a_2) = 0.444 - 0.444 = 0$
  - According to classification error, the best split is a_1.

2.

- Decision tree
  - X:

	X	
	0	1
C1	60	40
C2	60	40

$$Classification\ error(X) = \frac{60}{200} + \frac{40}{200} = 0.5$$

  - Y:

	Y	
	0	1
C1	40	60
C2	60	40

$$Classification\ error(Y) = \frac{40}{200} + \frac{40}{200} = 0.4$$

  - Z:

	Z	
	0	1
C1	30	70
C2	70	30

$$Classification\ error(Z) = \frac{30}{200} + \frac{30}{200} = 0.3$$

- Root node = Z
  - Z=0

	X		Y	
	0	1	0	1
C1	15	15	15	15
C2	45	25	45	25

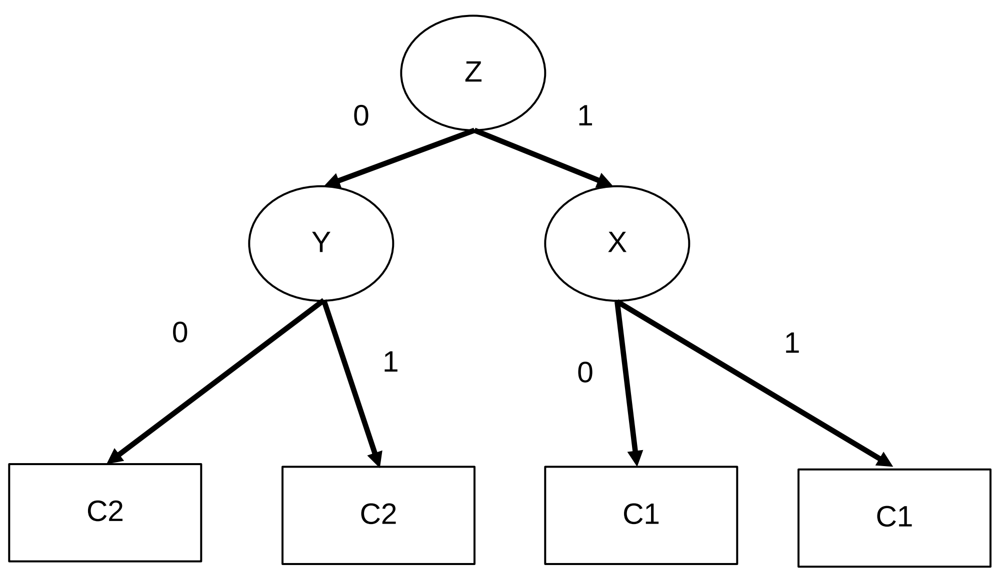
$$Classification\ error(Z_0, X) = \frac{15}{100} + \frac{15}{100} = 0.3$$
$$Classification\ error(Z_0, Y) = \frac{15}{100} + \frac{15}{100} = 0.3$$

  - Z=1

	X		Y	
	0	1	0	1
C1	45	25	25	45
C2	15	15	15	15

$$Classification\ error(Z_1, X) = \frac{15}{100} + \frac{15}{100} = 0.3$$
$$Classification\ error(Z_1, Y) = \frac{15}{100} + \frac{15}{100} = 0.3$$

- Since for both cases, the errors are the same, we can pick either X or Y for the child node. The second split seems redundant as the class decision depends mostly on Z.



- Error rate

  Since the target class depends on Z attribute, the error rate can be obtained by simply calculating the probability of the minority class of Z.

  $Error\ rate = \frac{30}{200} + \frac{30}{200} = 0.3$

- Feature importance

  First split(Z): $Impurity\ reduction = 0.5 - 0.3 = 0.2$

  Frist split(Y): $Impurity\ reduction = 0.5 - 0.4 = 0.1$

  First split(Z): $Impurity\ reduction = 0.5 - 0.5 = 0$

  Second split (Z=0, Y): $Impurity\ reduction = 0.3 - 0.3 = 0$

  Second split (Z=1, Y): $Impurity\ reduction = 0.3 - 0.3 = 0$

  Second split (Z=0, X): $Impurity\ reduction = 0.3 - 0.3 = 0$

  Second split (Z=1, X): $Impurity\ reduction = 0.3 - 0.3 = 0$

  Feature importance ranking = Z,Y,X

3. $k = 4$

   $N_{train} = 10$

   a) Generalization error rate using optimistic approach

      $err_{gen}(T) = err(T) = \frac{5}{10} = 0.5$

   b) Generalization error rate using pessimistic approach

      $err_{gen}(T) = err(T) + 0.5\left(\frac{k}{N_{train}}\right) = 0.5 + 0.5\left(\frac{4}{10}\right) = 0.7$

   c) Generalization error using validation set

      $err_{val}(T) = \frac{2}{5} = 0.4$

   d) Accuracy

Instance	A	B	C	Class (actual)	Result (predicted)	Result
16	0	1	0	+	-	FN
17	1	0	0	+	+	TP
18	1	1	1	-	-	TN
19	1	0	1	+	-	FN
20	1	1	1	-	-	TN
21	0	0	1	-	+	FP
22	1	1	0	+	+	TP
23	0	0	1	+	+	TP

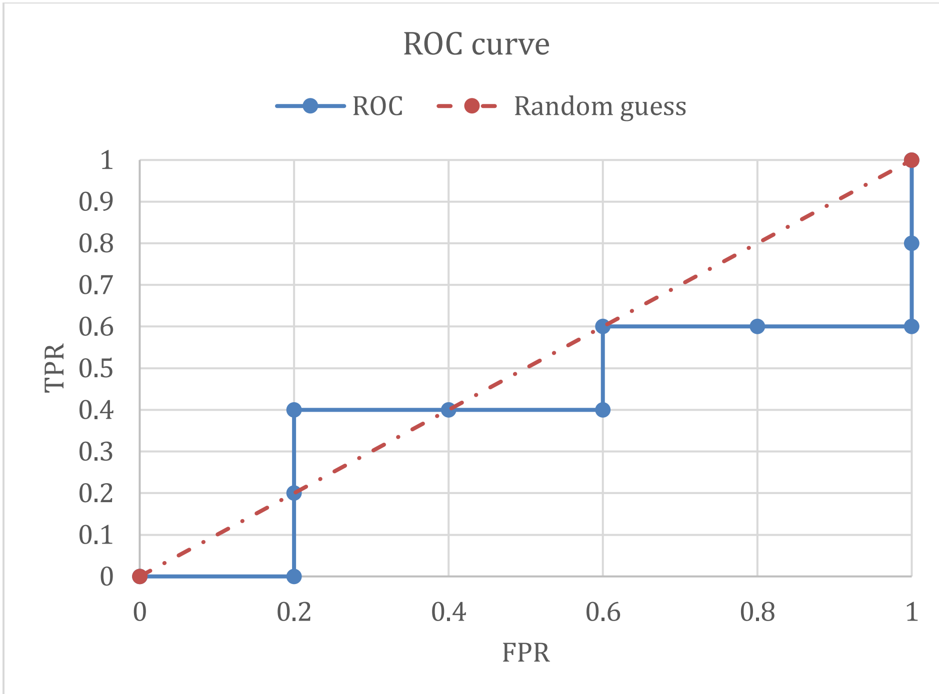
$TP = 3$
$TN = 2$
$FP = 1$
$FN = 2$
$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} = \frac{5}{8} = 0.625$$

e) $Precision = \frac{TP}{TP+FP} = \frac{3}{3+1} = \frac{3}{4} = 0.75$
$$Recall = \frac{TP}{TP + FN} = \frac{3}{3+2} = \frac{3}{5} = 0.6$$
$$F1 - measure = \frac{2rp}{r+p} = \frac{2(0.6)(0.75)}{0.6 + 0.75} = 0.667$$

4. ROC curve
   - Sort

Instance	True class	P(+|A)
3	-	0.68
1	+	0.61
5	+	0.45
7	-	0.38
4	-	0.31
6	+	0.09
8	-	0.05
10	-	0.04
1	+	0.03
9	+	0.01

Class	+	+	-	-	+	-	-	+	+	-	
	0.01	0.03	0.04	0.05	0.09	0.31	0.38	0.45	0.61	0.68	1
TP	5	4	3	3	3	2	2	2	1	0	0
FP	5	5	5	4	3	3	2	1	1	1	0
TN	0	0	0	1	2	2	3	4	4	4	5
FN	0	1	2	2	2	3	3	3	4	5	5
TPR	1	0.8	0.6	0.6	0.6	0.4	0.4	0.4	0.2	0	0
FPR	1	1	1	0.8	0.6	0.6	0.4	0.2	0.2	0.2	0



5. Github repo: https://github.com/fidelisprasetyo/CS5990/tree/hw3
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